

ARAB YOUTH SPACE HACKATHON 2026 CHALLENGE 813 PROOF OF CONCEPT

Madar Oman

Flood evidence through cloud — with a stated noise floor.

Team T0045 | Theme 01 — Climate Adaptation and Disaster Recovery | Sultanate of Oman

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THE PROBLEM

Oman's floods arrive under cloud and radar has a trap here

2 weeks

After the flash floods of 14–15 April 2024, the first usable Sentinel-2 scene came on 29 April, with 17–75 % cloud.

Dry sand looks like water

At C-band, fine dry sand is as smooth and dark as standing water. A simple threshold over northern Oman reports floods that never happened.

No stated confidence

Flood maps are published without saying how much of the mapped water would also appear on a dry day.

Who is affected: residents of the Al Batinah coast and the wadis of Ash Sharqiyah; civil defence, municipalities and road authorities deciding where to go first.

BUSINESS USE CASE

The duty analyst needs to know: is this map an event, or noise?

End user and decision

- Operations rooms in civil defence and municipalities: which roads and neighbourhoods first.
- Engineers and loss adjusters: which repairs and claims the evidence supports.
- Today they rely on field calls, late optical images and global products not tuned for desert terrain.

What we offer

- Subscription — national operations centre and analysis seats for public bodies.
- Per-event reports — for insurers and consultancies.
- On-premise deployment — for agencies with data-residency rules.
- Core data is open; our costs are compute and people.

THE PLATFORM AT A GLANCE

What it is for, how to use it, and how it will predict

Purpose

- Turn satellite and weather data into evidence a decision-maker in Oman can act on.
- Three answers every time: where the hazard is, how sure we are, and what is exposed to it.
- One place for floods, drought and coastal change, in Arabic and English.

How you benefit

- Before: all 61 wilayats ranked by risk, so attention goes where it is needed.
- Through cloud: radar images the ground on its next pass, every few days.
- After: roads and built land inside the water, and how fast it recedes — evidence for recovery and claims.
- Long term: shoreline and farm-area change, year by year.

How to use it

- Step 1 — open the operations centre and pick a wilayat — or draw or upload your own area.
- Step 2 — choose an indicator and a period; for a flood, give the event date.
- Step 3 — start. The platform fetches the imagery and shows progress.
- Step 4 — read the map, dashboard and report; download the PDF or the GIS layers.

How it will predict

- Today: a risk score per wilayat from open weather-model forecasts, rule-based, every factor visible.
- Next (planned, not built yet): pair each mapped flood with the rain that caused it, so a rain forecast shows where water stood before.
- Then: drought outlook from vegetation trends and seasonal forecasts.
- Always labelled a forecast; official warnings stay with the national centre.

DATA

Open Earth-observation data, each for one job

DATASET	PROVIDER	USED FOR
Sentinel-1 RTC, VV, 20 m	ESA / Copernicus	Flood extent through cloud; baseline per viewing geometry
Sentinel-2 L2A, 10 m	ESA / Copernicus	Vegetation and moisture indices; shoreline year by year
Copernicus DEM GLO-30	ESA / Copernicus	Slopes over 5° excluded (radar shadow)
ESA WorldCover 10 m	ESA	Permanent water excluded; built-up exposure
ERA5	ECMWF / Copernicus	Rainfall context for drought

Hyperspectral stated plainly

Our headline result is a radar result and uses no hyperspectral data. We restrict every spectral index in code to Satellite 813's 400–1700 nm range, so an index 813 cannot compute is refused, not borrowed from another sensor. We have located one EnMAP before/after pair over Ibra / Al Qabil (fifteen months apart: geomorphic change only). We have not processed 813 data yet.

APPROACH

Flood is water now that is not water normally — and more than an ordinary day shows

01 Read
10 Sentinel-1 VV passes, one viewing geometry, 20 m grid

02 Baseline
Median of 5 earlier passes: what the ground normally is

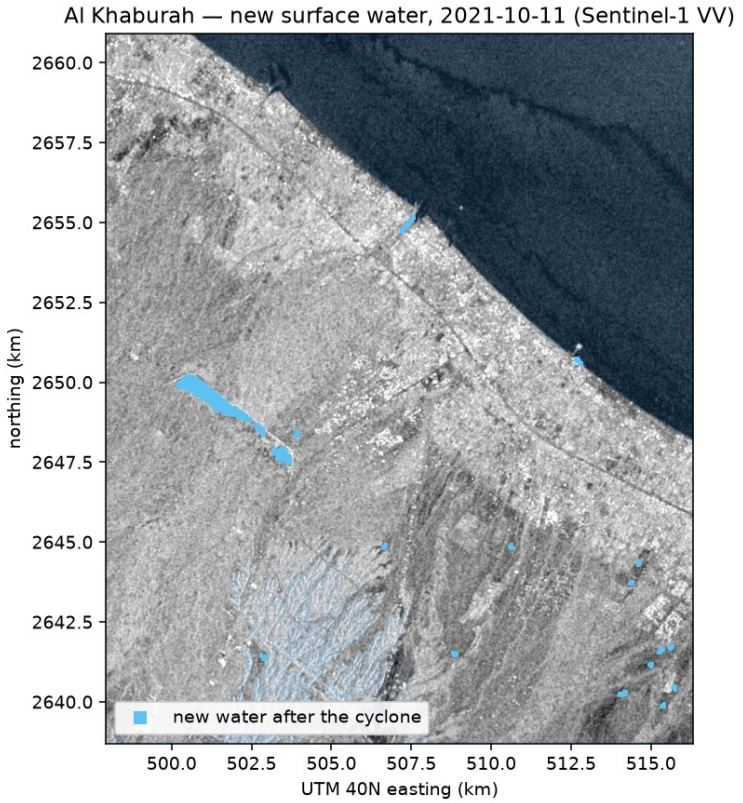
03 Mask
VV < −16 dB now, not in baseline, not steep, not permanent water, ≥ 0.5 ha

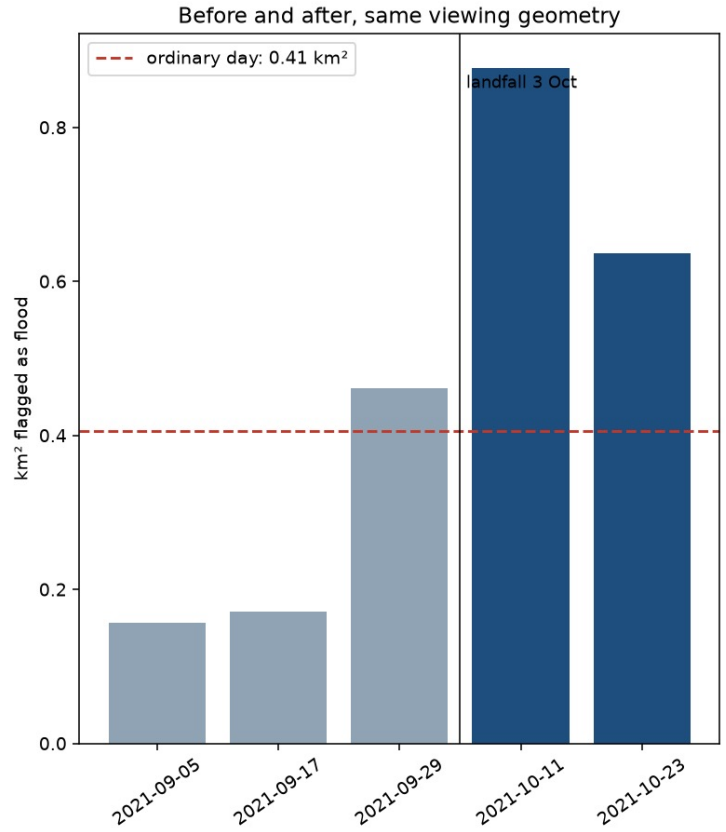
04 Split
Passes before landfall are ordinary days

05 Subtract
Remove every cell flagged before the event → new water

06 Null test
Same subtraction on a dry day = noise floor. Declare only above it

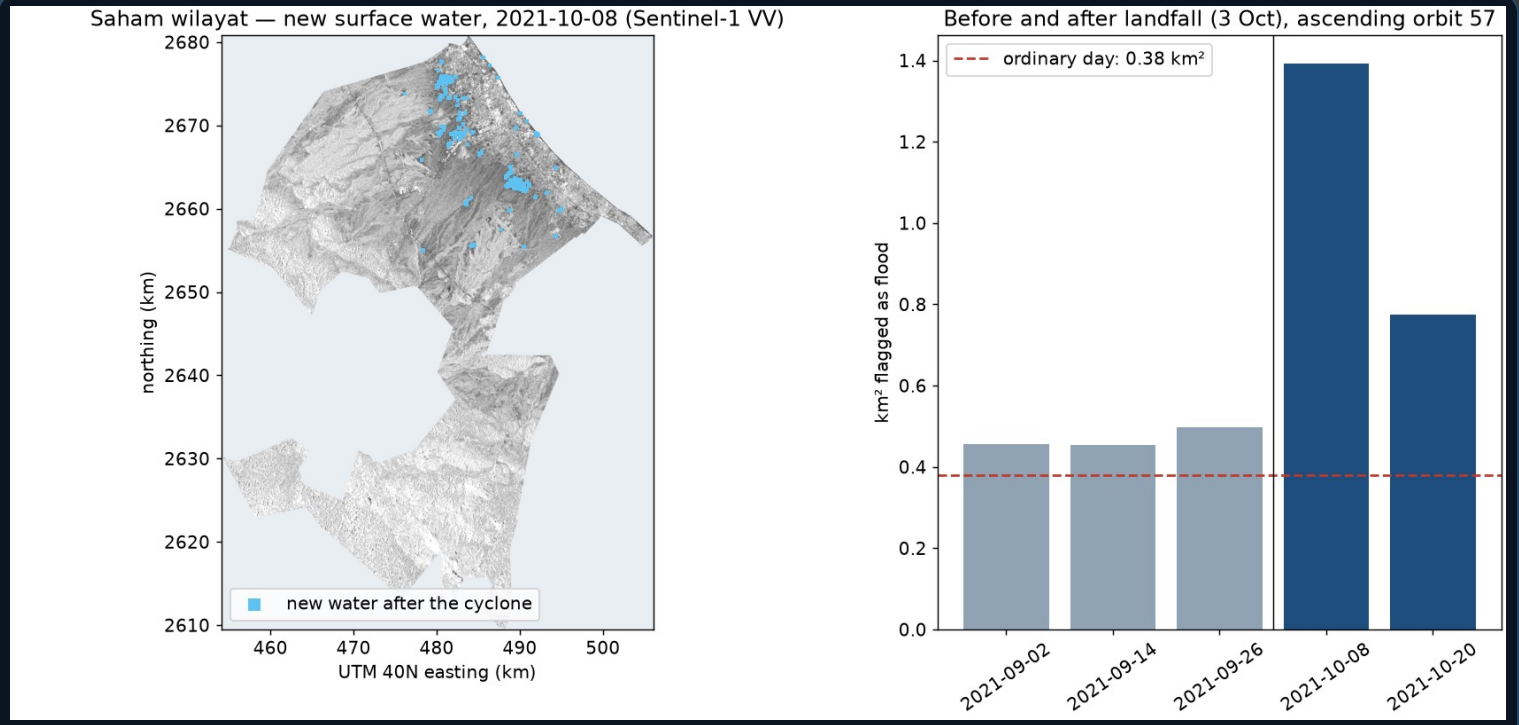
Al Khaburah after Cyclone Shaheen: 0.88 km² of new water, against 0.41 km² on an ordinary day





CASE 2 · FLOOD · SAHAM WILAYAT

Saham: 1.40 km² of new water against 0.37 km² — with roads and built land inside it



What the report states

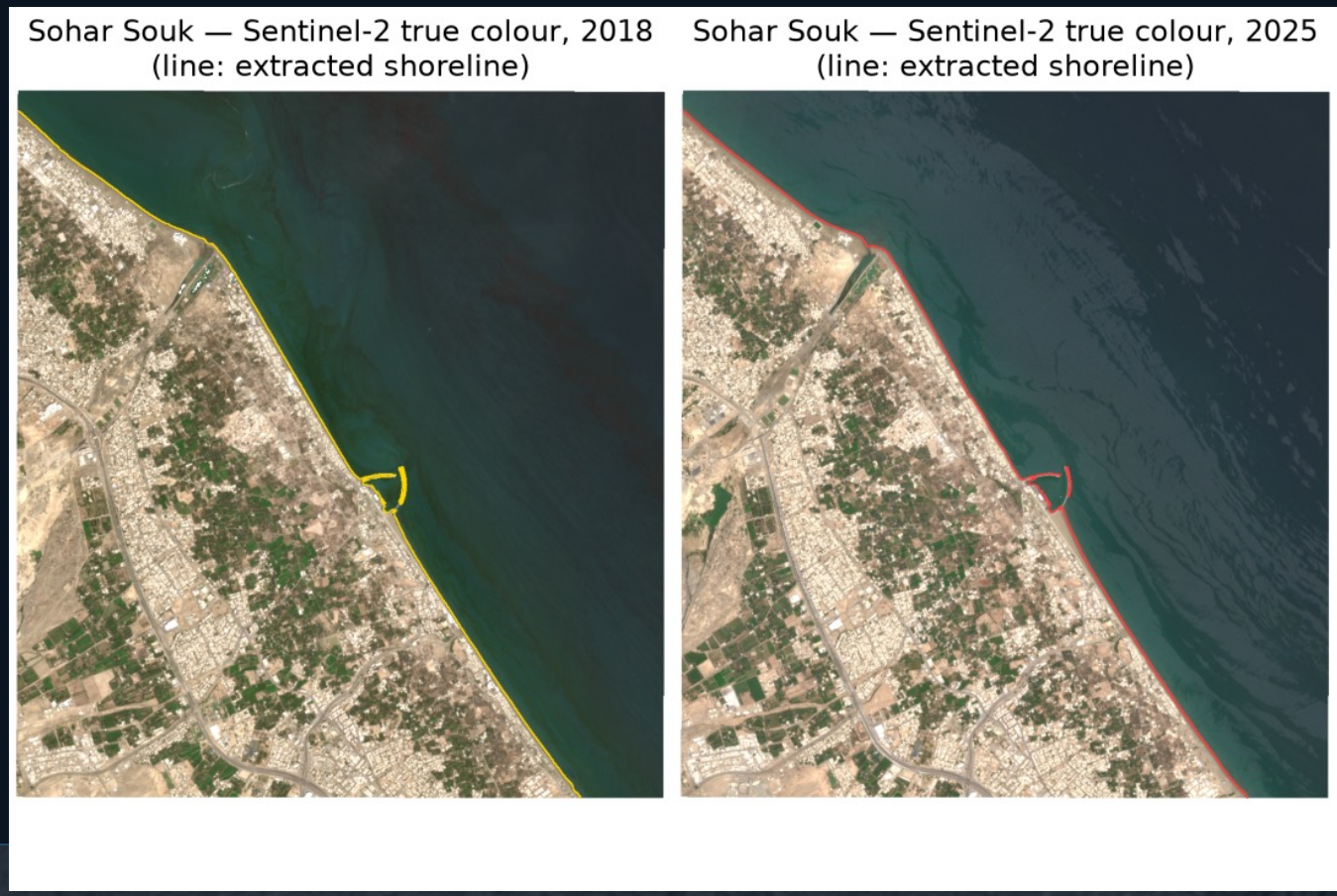
1.40 km² of new water on 8 Oct 2021, five days after landfall

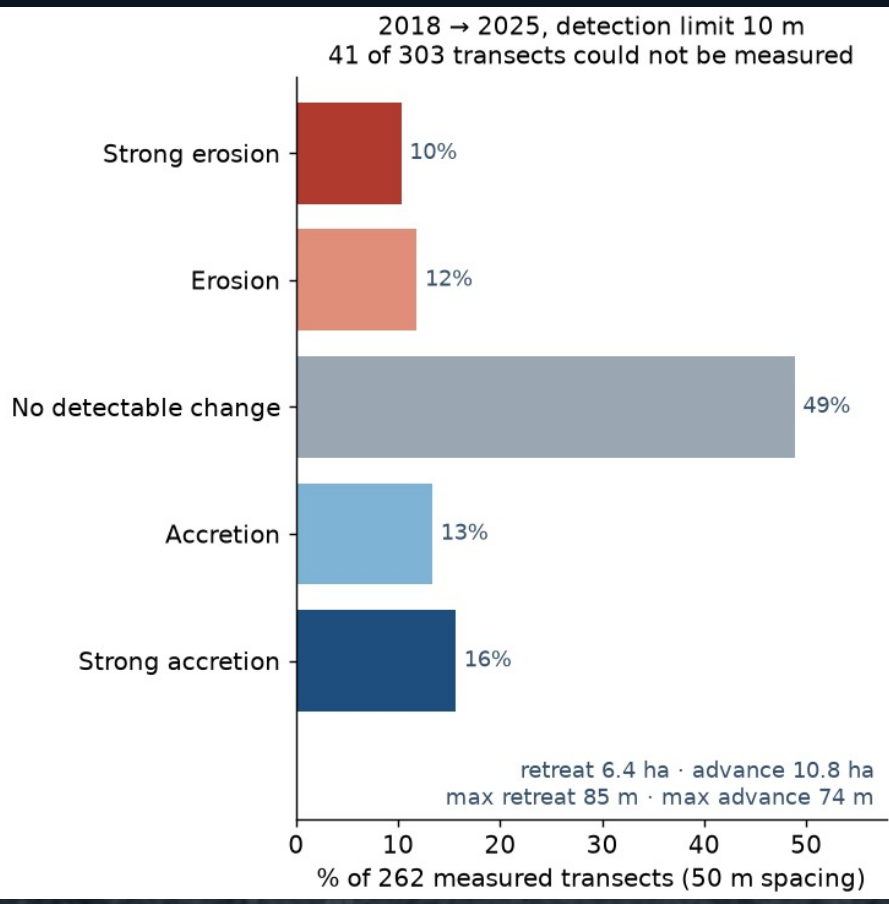
0.37 km² on an ordinary day, same method

2.96 km of mapped road reached

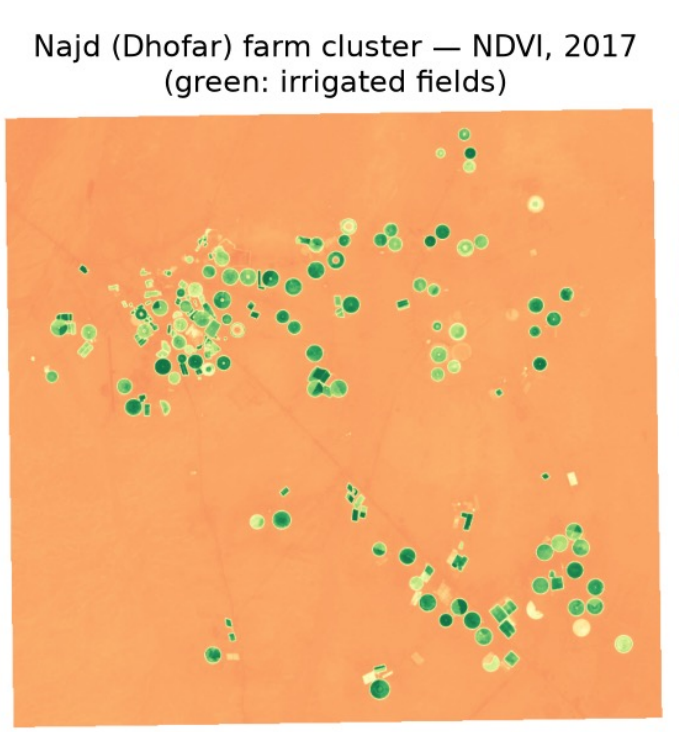
0.26 km² of built land reached — a lower bound

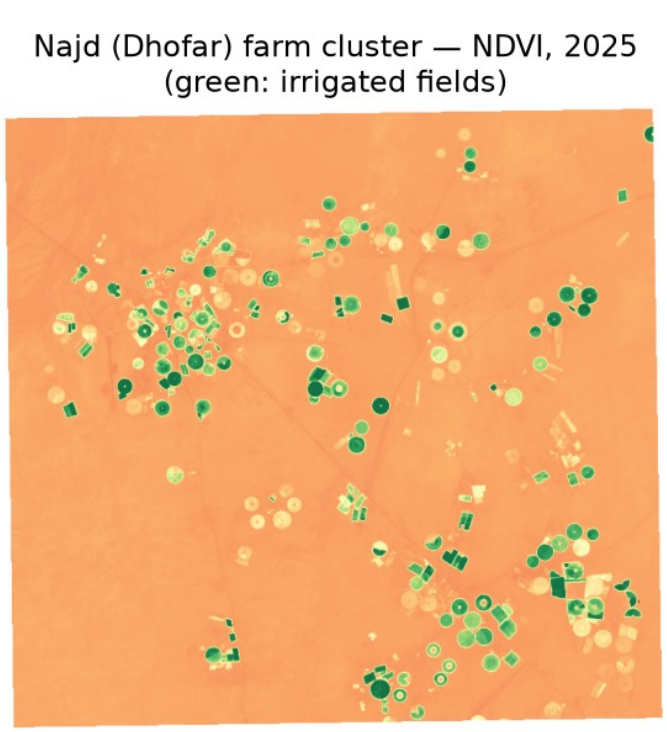
Sohar Souk shoreline, 2018 → 2025: 49% of measured transects show no detectable change

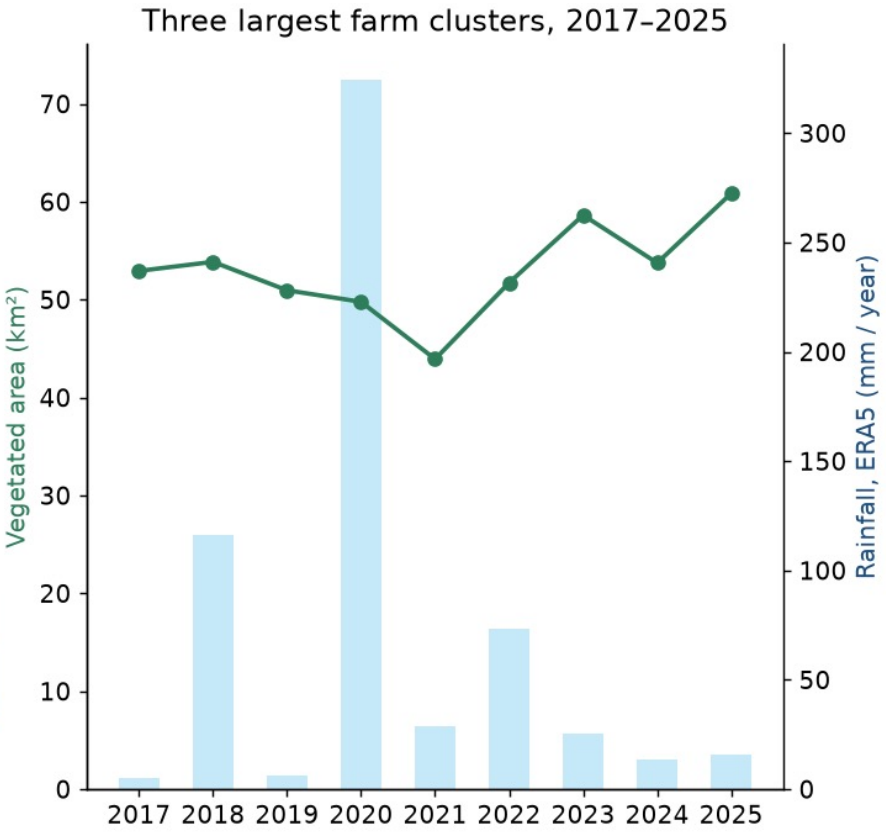




Najd farms, 2017 → 2025: vegetated area between 44 and 61 km², tracked against rainfall







Sentinel-2 NDVI and ERA5 rainfall for the three largest farm clusters. Lowest vegetated area 43.9 km² in 2021, highest 60.9 km² in 2025. In this record the farmed area does not rise and fall with rainfall.

THE PLATFORM

One platform behind every case: draw an area, choose an indicator, get a sourced report



What is running today

National risk table: 61 wilayats, ten hazards, every score decomposable into its factors.

Analysis workspace: flood, drought, coastal change and spectral indices on any drawn or uploaded area.

Reports in Arabic and English, HTML and PDF, with sources and limits.

Thirteen data feeds; twelve need no credentials.

Basemap shown: Sentinel-2 cloudless by EOX.

INDICATORS YOU CAN USE

22 analyses in eight families — 21 are available to run on an area you draw

Water and climate

- Flood extent, receding and impact (radar)
- Water bodies
- Water-quality indices
- Drought
- Moisture indicators
- Oil spill detection (radar)
- Vessel detection (radar)

Vegetation and agriculture

- Vegetation cover
- Vegetation health
- Tree cover and fragmentation (SDG 15.1.1)
- Agricultural land
- Mangroves

Land and coast

- Land-cover change
- Land degradation and bare soil
- Terrain and land suitability
- Ground subsidence — susceptibility screen
- Shoreline change
- Beach erosion

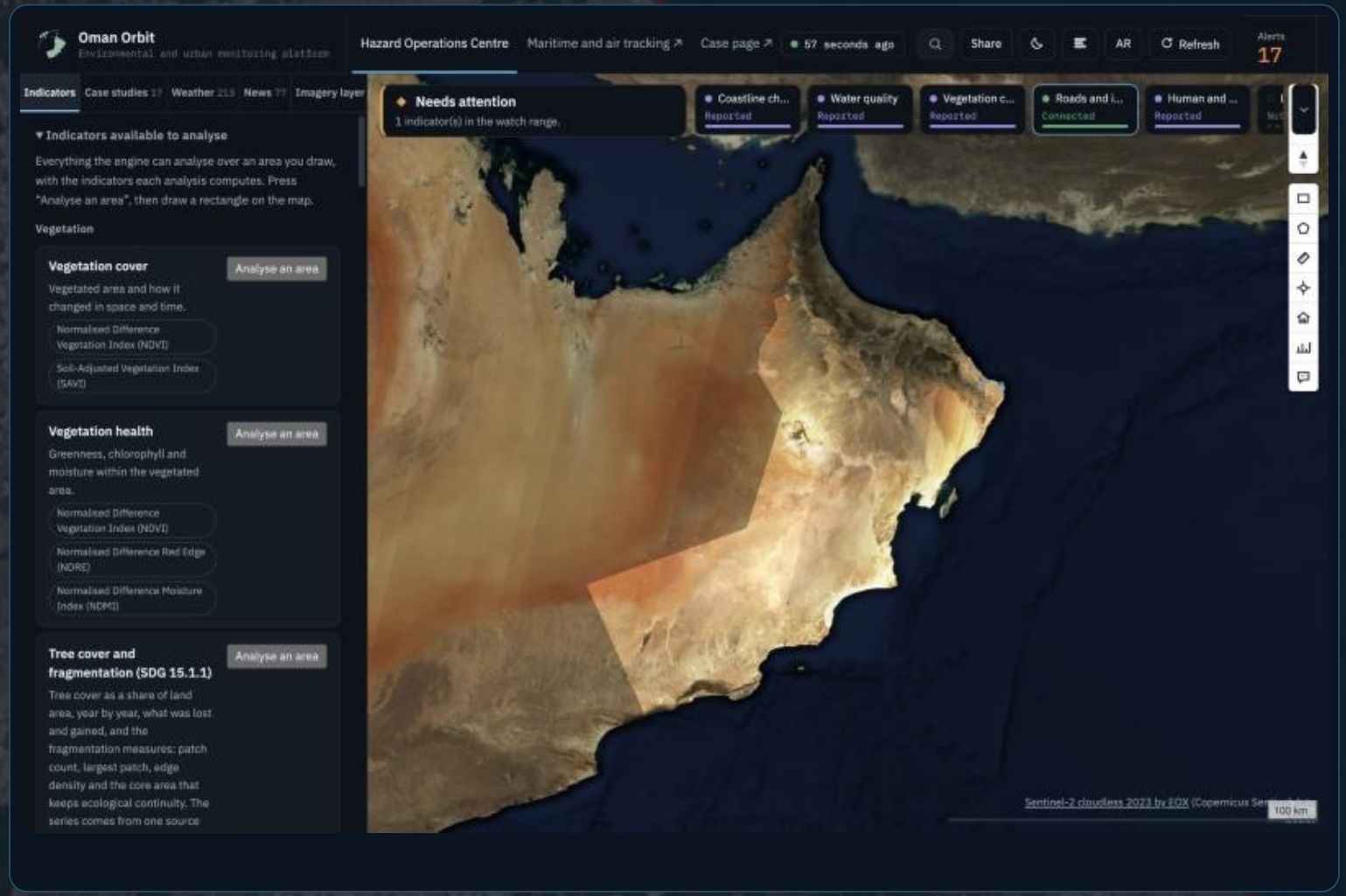
Urban and infrastructure

- Urban expansion
- Paved and unpaved roads
- Pipeline encroachment
- Urban heat — listed, not available yet

Spectral indices behind them: NDVI, SAVI, NDRE, NDMI, MNDWI, NDBI, BSI, NDCI, NDTI, VARI — plus radar backscatter, elevation and slope, ERA5 rainfall and soil moisture.

THE PLATFORM · SCREENSHOT

Step 1 — choose an indicator from the catalogue



Indicators tab

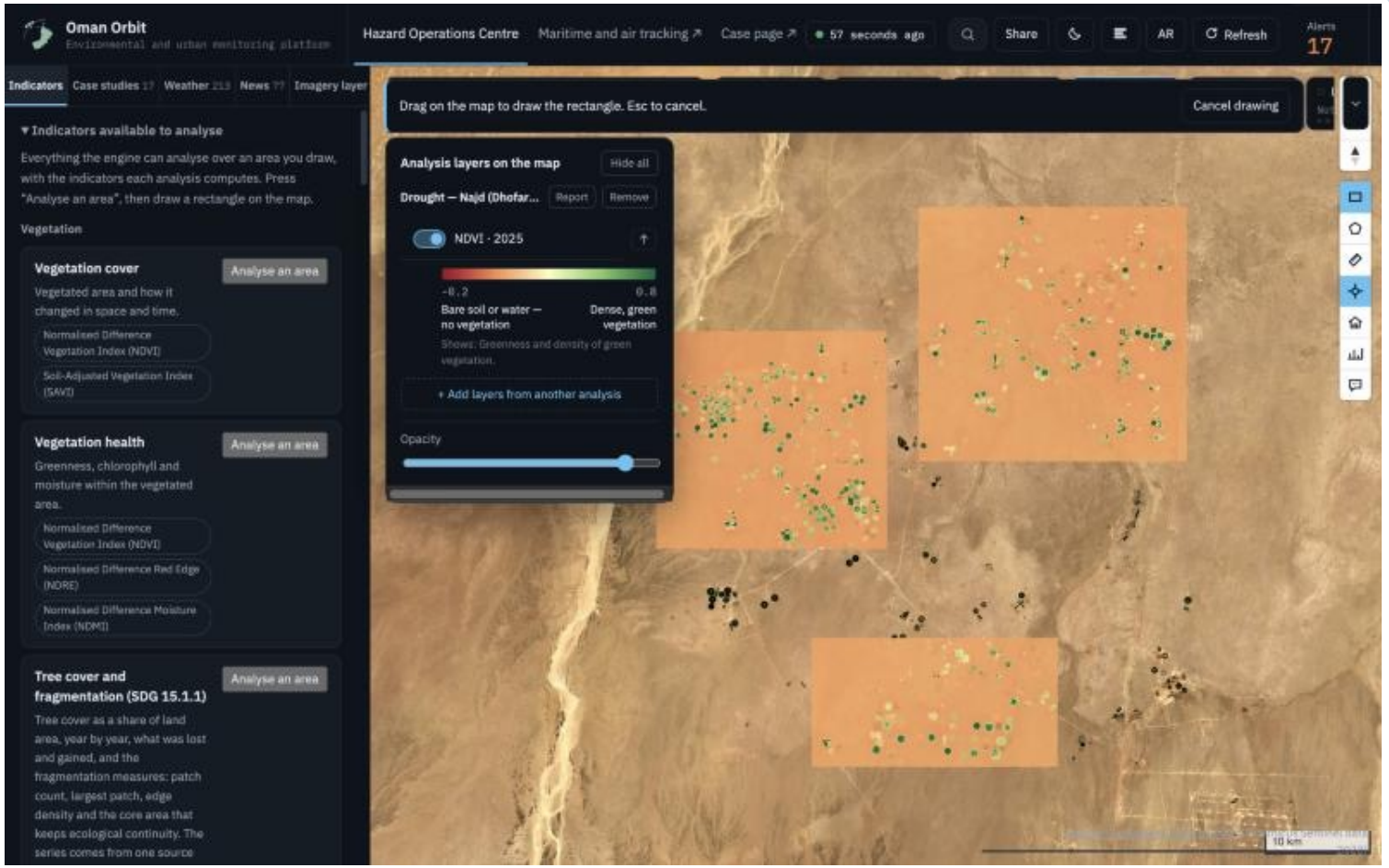
Every analysis is listed with what it measures and the indices it computes.

Press "Analyse an area" beside the one you need.

The top bar shows live readings: alerts, events, highest risk, sources.

Basemap: Sentinel-2 cloudless by EOX.

Step 2 — draw the area, then stack result layers on the map



On the map

Drag to draw a rectangle, draw a polygon, or upload a KML, Shapefile or GeoJSON.

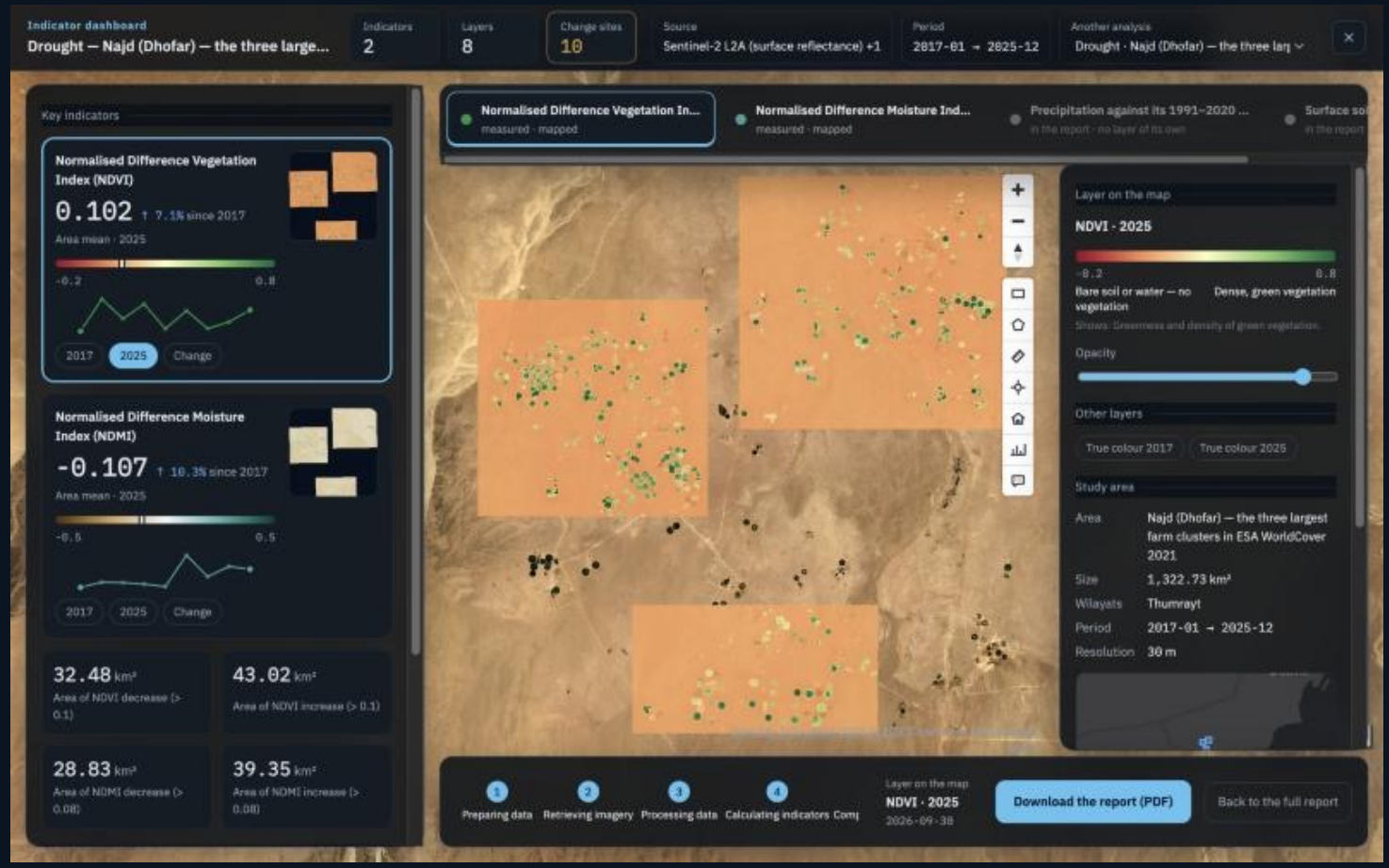
Each finished analysis adds layers you can switch on, reorder and fade.

Layers from several analyses can sit on the same map.

Shown: NDVI 2025 over the Najd farm clusters.

THE PLATFORM · SCREENSHOT

Step 3 — read the indicator dashboard and download the report



Indicator dashboard for the Najd drought study: key indicators with their trend since 2017, the mapped layer with its legend, the study area, and the report as PDF.

VALIDATION AND LIMITATIONS

We report "detected", "not distinguishable" and "cannot tell"

AREA, CYCLONE SHAHEEN 2021	NEW WATER	ORDINARY DAY	READING
Al Khaburah town and coastal plain	0.88 km²	0.41 km²	Detected, in both viewing geometries
Saham wilayat	1.40 km²	0.37 km²	Detected
Al Musannah wilayat (landfall)	2.17 km²	2.03 km²	**Not distinguishable from noise**
As Suwayq, Sohar	—	—	Too few usable passes; analysis refused

How we validated

The null test, and agreement between two independent viewing geometries. That is internal consistency — not ground truth.

Where it breaks

First usable passes came five to eight days after landfall; water between buildings is invisible to radar; rough or vegetated water is missed; no field survey of this event.

IMPACT

Evidence that arrives while it still changes decisions

Societal

A defensible answer to "which roads and neighbourhoods are inside the water" in the days when cloud blocks every optical satellite.

Economic

Evidence for recovery spending, infrastructure siting and insurance assessment — with the uncertainty stated.

Strategic

SDG 11 and SDG 13. Supports, and does not duplicate, Oman's national early-warning mandate. Methods use global open data and transfer to any Arab country with wadis and sabkhas.

NEXT STEPS FOR INCUBATION

From a reproducible result to a deployed service

1

Independent validation

Measure one flood map against a third-party product and reported impact sites.

2

Hyperspectral

First EnMAP / Satellite 813 indicator selectable and runnable in the workspace.

3

GIQ

Publish the analysis service on the GIQ marketplace.

4

Pilot

One Omani public body, one past event it knows well, judged against its own records.